Name: ____________________

STATISTICS, (MEAN, MEDIAN, MODE AND RANGE)

Date: ______________

LO: I can calculate and interpret the mean, median, mode and range for a data set.

Averages and range

The **mean** = total of values ÷ number of values. The **median** = middle value when ordered. The **mode** = most common value.

The **range** = highest value - lowest value. It measures the **spread** of the data.

Quick examples

🟢	Mean of 2, 4, 6, 8 = 20 ÷ 4 = 5	sum ÷ count
🟢	Median of 1, 3, 5, 7, 9 = 5	middle value
🟢	Mode of 2, 3, 3, 4, 5 = 3	most frequent
🔴	Median of 2, 4, 6, 8 = 6	mean of 4 and 6 = 5
🔴	Range of 3, 7, 1, 9 = 9	range = 9 - 1 = 8

Key idea

Mean = sum ÷ n. Median = middle value (order first). Mode = most common. Range = max - min.

$$\text{Mean} = \Sigma x \div n$$

Sum of all values divided by the count

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – calculating the mean

Find the mean of 5, 8, 3, 12, 7.

Sum = 5 + 8 + 3 + 12 + 7 = 35

Count = 5

Mean = 35 ÷ 5 = 7

Mean = 7

Add all values then divide by how many there are.

Example 2 – finding the median

Find the median of 9, 3, 7, 1, 5, 8, 2.

Order: 1, 2, 3, 5, 7, 8, 9

7 values, middle is the 4th

Median = 5

Median = 5

With an odd number of values, pick the middle one.

Example 3 – even number of values

Find the median of 4, 7, 2, 9, 6, 8.

Order: 2, 4, 6, 7, 8, 9

6 values: middle is between 3rd and 4th

Median = (6 + 7) ÷ 2 = 6.5

Median = 6.5

With an even number, average the two middle values.

Example 4 – from frequency table

Scores: 1(f=3), 2(f=5), 3(f=7), 4(f=4), 5(f=1). Find the mean.

Sum = 1×3 + 2×5 + 3×7 + 4×4 + 5×1

= 3+10+21+16+5 = 55

Total frequency = 20

Mean = 55 ÷ 20 = 2.75

Mean = 2.75

Multiply each value by its frequency, sum, then divide by total frequency.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding the mean
Calculate the mean of each data set.

a) 4, 6, 8, 10, 12

b) 3, 7, 5, 9, 1

c) 15, 20, 25, 30, 10

d) 2.5, 3.5, 4, 5, 5

2. Finding the median
Find the median of each data set.

a) 3, 7, 1, 9, 5

b) 12, 8, 15, 3, 10, 6

c) 2, 2, 5, 7, 8, 9, 11

d) 4, 6, 8, 10

3. Mode and range
Find the mode and range of each data set.

a) 3, 5, 3, 7, 3, 9

b) 1, 4, 4, 6, 6, 6, 8

c) 10, 20, 30, 20, 40

d) 5, 5, 8, 8, 10

4. From frequency tables
Find the mean from each frequency table.

a) Score 1(f=2), 2(f=5), 3(f=3). Mean?

b) Goals 0(f=4), 1(f=6), 2(f=5), 3(f=5). Mean?

c) Rating 1(f=3), 2(f=4), 3(f=8), 4(f=5). Mean?

d) Value 5(f=2), 10(f=3), 15(f=4), 20(f=1). Mean?

5. Problem solving
Solve each problem about averages.

e) Mean of 6 values is 5. One value

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) The mean is found by adding all values and dividing by the count ☐

Correct answer: __________

Reason:

b) The median of 2, 4, 6, 8 is 6 ☐

Correct answer: __________

Reason: __________________

c) A data set can have more than one mode ☐

Correct answer: __________

Reason: __________________

d) The range is the biggest number in the data set ☐

Correct answer: __________

Reason: __________________

e) You must order data before finding the median ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The mean of 10 test scores is **72**. One student scored **45** and retakes, scoring **78**. What is the new mean of the 10 scores?

Expression: ______________________________

Simplified: ______________________________

b) A set of 7 numbers has median **12**, mode **15**, and range **11**. The smallest number is **7**. Find a possible set of 7 numbers that satisfies all conditions.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!